

Agenda

1. Friday Feedback



2. Section 9.1

3. Section 9.2



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Solutions to Differential Equations

We call a function a solution if it satisfies the given differential equation.

Example: Show that $y = 2e^{-x} + x - 1$ is a solution to $y' = x - y$

Step 1: find y' : $y' = -2e^{-x} + 1$

Step 2: start with one side & simplify until it looks like the other.

$$\begin{aligned} \text{RHS} &= x - y = x - (2e^{-x} + x - 1) \\ &= x - 2e^{-x} - x + 1 = -2e^{-x} + 1 = y' = \text{LHS} \quad \checkmark \end{aligned}$$

Activity: Show that $y = 4(x^2 - 1)^{-1/2}$ is a solution to $(x^2 - 1)y' + xy = 0$.

$$y' = -2(x^2 - 1)^{-3/2} \cdot 2x$$

$$\begin{aligned} \text{LHS} &= (x^2 - 1)y' + xy = (x^2 - 1)(-4x)(x^2 - 1)^{-3/2} + (4x)(x^2 - 1)^{-1/2} \\ &= 0 = \text{RHS} \quad \checkmark \end{aligned}$$

Example: Find all values of k such that $y = \sin(kt)$ is a solution to $y'' + 16y = 0$

Step 1: find y'' $y = \sin(kt)$ $y' = k \cos(kt)$ $y'' = -k^2 \sin(kt)$

Step 2: substitute and solve:

$$\begin{aligned} y'' + 16y &= 0 \\ -k^2 \sin(kt) + 16 \sin(kt) &= 0 \\ (\sin(kt))(16 - k^2) &= 0 \end{aligned}$$

~~$\sin(kt)$~~ or
 $16 - k^2 = 0$
 $\pm k = 4$

Activity: Find the values of n so that $y = e^{nt}$ is a solution to $y' = 6y - y''$ $y' = ne^{nt}$ $y'' = n^2 e^{nt}$

$$\begin{aligned} ne^{nt} &= 6e^{nt} - n^2 e^{nt} \\ n^2 e^{nt} - ne^{nt} - 6e^{nt} &= 0 \end{aligned}$$

$$0 = (n^2 - n - 6)e^{nt}$$

$$(n+6)(n-1)$$

$$n = -6, n = 1$$

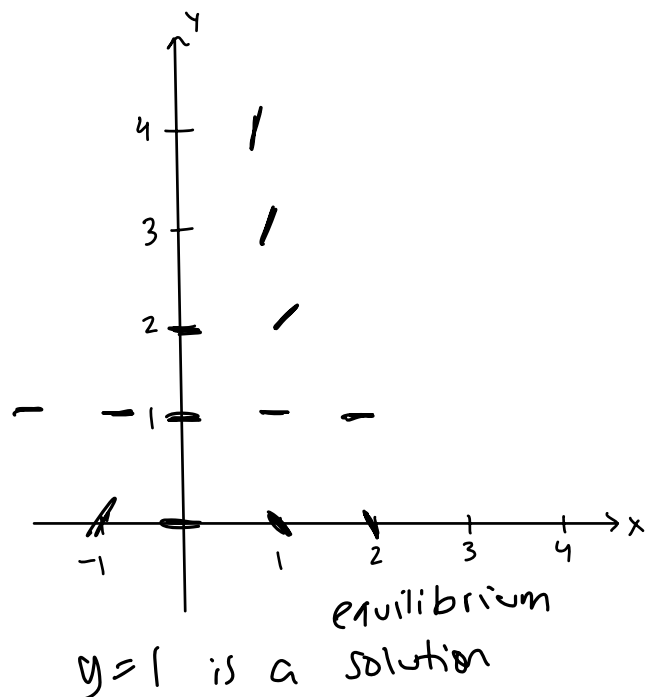
Section 9.2 - Direction Fields and Euler's Method

It is often impossible to solve for the solutions of a differential equation by finding a solution directly.

Direction Fields

Example: $\frac{dy}{dx} = yx - x$

x, y	$\frac{dy}{dx} = yx - x$
$(0, 0)$	0
$(1, 0)$	-1
$(2, 0)$	-2
$(0, 1)$	0
$(0, 2)$	0
$(1, 1)$	0
$(2, 1)$	0



Definition: A direction field (or slope field) of a differential equation is a graph of short slope lines that correspond to the differential activity at a particular point.